

ONEDRAFT

CAD-AI — Aria Powered

OneDraft–ArcGIS Repository Application Implementation Specifications

Version 1.0

Document: OneDraft–ArcGIS Repository Application Implementation Specification

System: OneDraft CAD-AI

Intelligence Layer: Aria

Document Class: Architectural & Implementation Stack Specification

Status: Implementation Specification

Predecessor: Global Geospatial Application, Repository & World-Model Implementation Architecture

Successor: To Be Determined by Implementation Requirements

1. PURPOSE

Stack 53.0 defines the executable implementation architecture required to connect the OneDraft platform to authorized ArcGIS repositories and services.

Stack 52.0 established the Global Geospatial Application, Repository & World-Model architecture.

Stack 53.0 converts that architecture into an implementation specification governing:

ArcGIS connectivity

Repository integration

GIS dataset registration

Spatial data ingestion

Spatial data normalization

Feature and layer management

Coordinate-reference management

Spatial synchronization

Project georeferencing

Authorized repository write-back

Spatial provenance

Version management

Conflict management

API integration

Job execution

Database persistence

Security

Testing

Deployment

The objective is to establish a concrete implementation boundary between the OneDraft application and external ArcGIS infrastructure while preserving OneDraft as an independent architectural and engineering platform.

2. IMPLEMENTATION OBJECTIVE

The objective of Stack 53.0 is to provide the executable system required to establish:

```
ONEDRAFT
↓
GEOSPATIAL APPLICATION
↓
ARCGIS ADAPTER
↓
ARCGIS SERVICES
↓
ARCGIS REPOSITORY
```

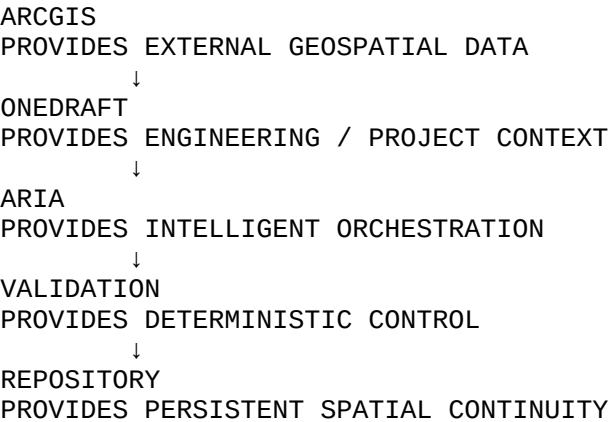
and the reverse information path:

```
ARCGIS REPOSITORY
↓
ARCGIS ADAPTER
↓
INGESTION
↓
NORMALIZATION
↓
VALIDATION
↓
ONEDRAFT SPATIAL MODEL
↓
PROJECT / SITE / STRUCTURE
```

The implementation shall support controlled bidirectional interaction without collapsing the distinction between external GIS information and canonical OneDraft project information.

3. IMPLEMENTATION PRINCIPLE

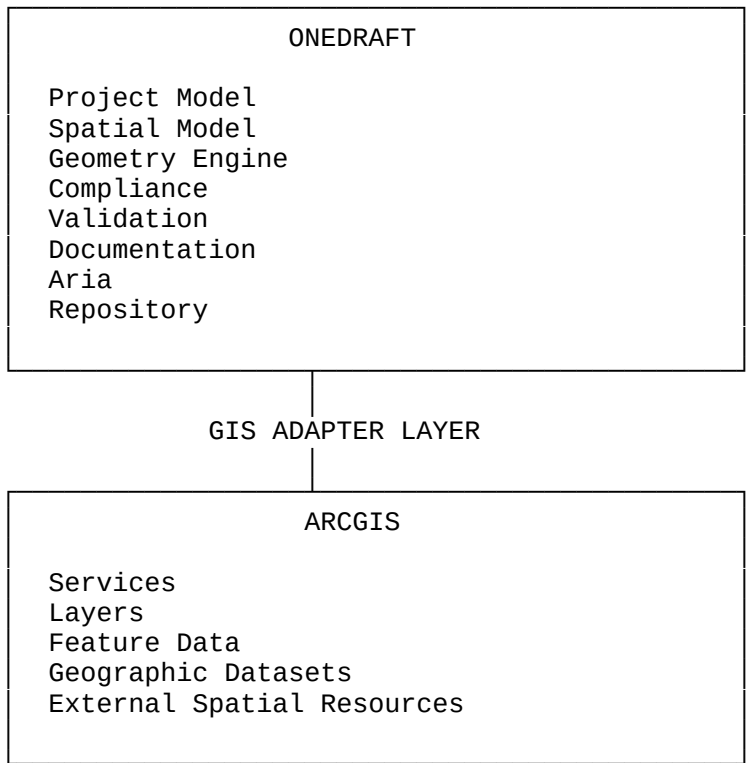
The primary implementation principle shall be:



No subsystem shall assume that another subsystem is authoritative outside its defined domain.

4. SYSTEM BOUNDARY

Stack 53.0 establishes the following implementation boundary:



The adapter layer shall isolate the OneDraft domain from provider-specific implementation details.

5. ARCGIS ADAPTER

The ArcGIS Adapter shall be the primary integration boundary between OneDraft and ArcGIS services.

The adapter shall provide normalized OneDraft interfaces for:

Dataset discovery

Layer discovery

Feature queries

Spatial queries

Metadata retrieval

Feature retrieval

Feature creation where authorized

Feature updates where authorized

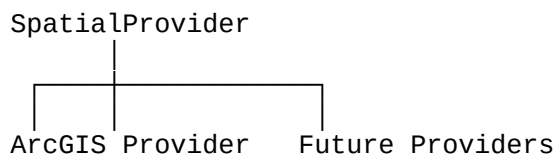
Synchronization

Service health

The application shall not distribute ArcGIS-specific logic throughout the OneDraft domain layer.

6. PROVIDER ABSTRACTION

The implementation shall use a provider abstraction.



ArcGIS shall be the first implementation.

Future providers shall be capable of connecting through the same spatial-provider contract.

7. ARCGIS SERVICE REGISTRY

OneDraft shall maintain an internal registry of authorized ArcGIS services.

A service registration shall include:

```
ArcGISService
├── id
├── provider
├── service_url
├── service_type
├── owner
├── access_policy
├── spatial_reference
├── status
├── metadata
└── synchronization_policy
```

Service credentials shall never be stored directly within ordinary domain records.

8. DATASET REGISTRY

The Dataset Registry shall maintain the canonical OneDraft record describing an external GIS dataset.

The registry shall contain:

Dataset identifier

Provider

Service

Layer

Dataset name

Source

Description

Spatial reference

Version

Acquisition timestamp

Last synchronization

License

Access classification

Provenance

Status

9. LAYER REGISTRY

Individual ArcGIS layers shall be represented independently of datasets.

DATASET
↓
SERVICE
↓
LAYER
↓
FEATURES

Layer metadata shall include:

Layer identifier

Layer name

Geometry type

Spatial reference

Fields

Capabilities

Extent

Source

Version

The layer registry shall permit OneDraft to discover the structure of external geographic information without importing all underlying data.

10. FEATURE REPRESENTATION

External ArcGIS features shall be represented through a normalized spatial feature model.

```
SpatialFeature
├── external_id
├── geometry
├── attributes
├── source
├── dataset
├── layer
├── spatial_reference
├── timestamp
└── provenance
```

External identifiers shall remain available after normalization.

11. CANONICAL SPATIAL MODEL

ArcGIS feature data shall be converted into the canonical OneDraft spatial model before becoming part of project context.

```
ARCGIS FEATURE
  ↓
ADAPTER
  ↓
NORMALIZATION
  ↓
CANONICAL SPATIAL ENTITY
```

The canonical model shall remain provider-neutral.

12. SPATIAL REFERENCE IMPLEMENTATION

The integration shall explicitly identify the coordinate reference system associated with every imported geometry.

The implementation shall preserve:

Source CRS

Target CRS

Transformation

Units

Precision

Transformation timestamp

No geometry shall be silently transformed without recording its transformation context.

13. COORDINATE TRANSFORMATION SERVICE

The Coordinate Transformation Service shall perform all supported geographic transformations.

```
SOURCE GEOMETRY
  ↓
SOURCE CRS
  ↓
```

TRANSFORMATION
↓
TARGET CRS
↓
ONEDRAFT GEOMETRY

The service shall be independently testable.

14. ARCGIS DISCOVERY WORKFLOW

The implementation shall support controlled discovery.

SERVICE
↓
DATASET
↓
LAYER
↓
METADATA
↓
FEATURE QUERY

Discovery shall occur before bulk ingestion.

This permits OneDraft to determine what external information exists before determining what information should be imported.

15. DATASET INGESTION PIPELINE

The implementation shall provide:

DISCOVER
↓
REGISTER
↓
RETRIEVE
↓
PARSE
↓
VALIDATE
↓
NORMALIZE
↓
PROVENANCE
↓
PERSIST
↓
INDEX

Each stage shall produce explicit success or failure state.

16. FEATURE INGESTION

Feature ingestion shall process external geographic features individually or in controlled batches.

The ingestion service shall support:

Geometry

Attributes

External identifiers

Dataset identifiers

Layer identifiers

Spatial references

Source timestamps

Provenance

Batch operations shall be resumable where practical.

17. ATTRIBUTE MAPPING

External attributes shall be mapped through explicit schemas.

```
ARCGIS FIELD
  ↓
FIELD MAPPING
  ↓
ONEDRAFT ATTRIBUTE
```

Unknown fields shall not automatically be discarded.

Where appropriate, they shall remain available as source attributes.

18. SCHEMA MAPPING REGISTRY

The system shall maintain mappings between external and canonical schemas.

A mapping shall define:

Source field

Target field

Data type

Transformation

Validation

Required state

Version

Schema mappings shall be version controlled.

19. GEOMETRY NORMALIZATION

The implementation shall normalize supported geometry types.

Examples include:

Point

Multipoint

Line

Polyline

Polygon

Multipolygon

Geometry collections where supported

Normalization shall preserve geometric meaning.

20. GEOMETRY VALIDATION

Every imported geometry shall pass validation appropriate to its geometry type.

Validation shall detect:

Invalid coordinates

Malformed geometry

Self-intersection where applicable

Invalid rings

Unsupported geometry

Coordinate-reference mismatch

Topology problems

Invalid records shall enter explicit failure or quarantine state.

21. SPATIAL INDEXING

Imported spatial entities shall be spatially indexed.

The index shall support:

Point lookup

Bounding-box queries

Polygon intersection

Proximity

Containment

Nearest feature

Project context

Spatial indexing shall be implemented independently from the external ArcGIS service.

22. POSTGRESQL SPATIAL PERSISTENCE

The canonical spatial model shall be persisted within the OneDraft PostgreSQL architecture using an appropriate spatial extension and schema strategy.

Conceptually:

```
PostgreSQL
├── projects
├── sites
├── structures
├── spatial_entities
├── spatial_datasets
├── spatial_layers
├── spatial_relationships
├── spatial_versions
└── spatial_provenance
```

Spatial persistence shall remain integrated with the existing OneDraft transaction model.

23. EXTERNAL IDENTIFIER PRESERVATION

Every imported external record shall preserve its source identity.

The implementation shall support:

Provider
+
Service
+
Dataset
+
Layer
+
External Feature ID

This prevents duplicate records from being created during subsequent synchronization.

24. DEDUPLICATION

The ingestion service shall identify potential duplicate spatial records.

Deduplication may consider:

External identifier

Dataset

Layer

Geometry

Attributes

Temporal state

Duplicate detection shall not automatically merge records where the distinction is meaningful.

25. PROJECT GEOREFERENCING

OneDraft projects shall be georeferenced through an explicit project operation.

PROJECT
↓
SITE
↓
GEOREFERENCE
↓

SPATIAL REGISTRATION
↓
GLOBAL PROJECT INDEX

The operation shall persist the relationship between the project and geographic location.

26. SITE REGISTRATION

The Site Registration Service shall establish:

Site identifier

Project relationship

Boundary

Coordinate reference

Location

Jurisdiction

Existing context

Source

The site shall become the primary spatial anchor for the architectural project.

27. STRUCTURE REGISTRATION

Existing structures may be registered independently from OneDraft projects.

EXTERNAL STRUCTURE
↓
STRUCTURE RECORD
↓
SPATIAL INDEX
↓
OPTIONAL PROJECT LINK

A structure shall not be assumed to be a OneDraft project merely because it is represented spatially.

28. ARCGIS-TO-ONEDRAFT READ PATH

The canonical read path shall be:

ARCGIS

↓
SERVICE
↓
DATASET
↓
LAYER
↓
FEATURE QUERY
↓
ADAPTER
↓
NORMALIZATION
↓
VALIDATION
↓
SPATIAL MODEL
↓
PROJECT CONTEXT

This shall be the primary external-data ingestion path.

29. ONEDRAFT-TO-ARCGIS WRITE PATH

Where write access is authorized, the write path shall be:

ONEDRAFT
↓
VALIDATED PROJECT DATA
↓
DATA CLASSIFICATION
↓
AUTHORIZATION
↓
EXPORT MAPPING
↓
ARCGIS ADAPTER
↓
ARCGIS SERVICE
↓
AUTHORIZED REPOSITORY UPDATE

Write-back shall never occur solely because Aria generated or modified information.

30. WRITE-BACK POLICY

Repository write-back shall require:

Valid project state

Explicit data classification

Authorized destination

Schema compatibility

Validation

Provenance

Authorization

The system shall distinguish between:

LOCAL PROJECT STATE

and:

GLOBAL REPOSITORY STATE

31. ARCGIS EDITING MODEL

Where external ArcGIS editing is supported, OneDraft shall treat external edits as externally sourced state.

The system shall record:

External version

External timestamp

Source

Changed fields

Changed geometry

Synchronization event

The OneDraft canonical state shall not be silently overwritten.

32. SYNCHRONIZATION ENGINE

The synchronization engine shall reconcile external and internal states.

ARCGIS STATE



COMPARE



CHANGE DETECTION



VALIDATION

↓
RECONCILIATION
↓
ONEDRAFT STATE

The reverse process shall apply to authorized write-back.

33. CHANGE DETECTION

The system shall detect changes using available identifiers, timestamps, versions, hashes or deterministic comparisons.

Change states may include:

UNCHANGED
ADDED
MODIFIED
REMOVED
SUPERSEDED
CONFLICT

34. CONFLICT MANAGEMENT

Conflicts shall be represented explicitly.

ONEDRAFT STATE
+
ARCGIS STATE
↓
CONFLICT
↓
RESOLUTION POLICY
↓
RESOLVED STATE

Resolution may be:

Automatic

Rule-based

Human-reviewed

Deferred

No conflict shall be silently discarded.

35. PROVENANCE IMPLEMENTATION

Every external spatial entity shall maintain source provenance.

The provenance record shall identify:

Provider

Dataset

Layer

Feature

Acquisition

Transformation

Importer

Version

Timestamp

The lineage shall remain queryable.

36. HASHING

Where applicable, source representations shall receive deterministic hashes.

Hashes may be used to detect:

Unchanged data

Modified data

Duplicate data

Artifact changes

Synchronization changes

Hashes shall complement rather than replace authoritative source identifiers.

37. TEMPORAL VERSIONING

Spatial entities shall support temporal versions.

FEATURE

↓

VERSION 1
↓
VERSION 2
↓
VERSION 3
↓
CURRENT

Historical records shall remain distinguishable according to retention policy.

38. HISTORICAL DATA MODEL

Historical information shall not be treated as current information merely because it remains present in the repository.

Each temporal record shall include appropriate state information:

HISTORICAL
EXISTING
CURRENT
PROPOSED
FUTURE
SUPERSEDED

39. GLOBAL PLACEHOLDER IMPLEMENTATION

The implementation shall support incomplete geographic records.

PLACEHOLDER
↓
LOCATION
↓
IDENTIFIED FEATURE
↓
STRUCTURE
↓
PROJECT
↓
DETAILED MODEL

A placeholder may therefore exist before full engineering information is available.

40. GLOBAL REPOSITORY REGISTRATION

A OneDraft project shall be capable of receiving a global spatial identifier.

Conceptually:

OneDraft Project ID
+
Spatial ID
+
Site ID
+
Structure ID
↓
Global Registry Record

Identifiers shall remain stable through the project lifecycle.

41. GLOBAL PROJECT INDEX

The Global Project Index shall support geographic discovery of authorized OneDraft projects.

The index shall permit queries by:

Country

Region

City

Coordinates

Site

Project

Structure

Project state

Temporal state

42. GLOBAL STRUCTURE INDEX

The Global Structure Index shall provide geographic discovery of registered structures.

The index shall distinguish:

Existing

Historical

OneDraft

Proposed

Unknown

Superseded

Structure records shall retain their source provenance.

43. ARCGIS DATASET MIRRORING

Where authorized, selected ArcGIS datasets may be represented locally within OneDraft.

The local representation shall be a controlled mirror.

ARCGIS
↓
AUTHORIZED DATASET
↓
LOCAL MIRROR
↓
ONEDRAFT INDEX

Mirroring shall not imply ownership or authority over the external source.

44. CACHE ARCHITECTURE

The implementation may maintain cached spatial information for performance.

Cache records shall have:

Source

Timestamp

Expiration

Version

Hash

Cached information shall not be confused with canonical project state.

45. OFFLINE / DEGRADED OPERATION

Where practical, OneDraft shall remain operational when ArcGIS services are temporarily unavailable.

The system shall distinguish:

ONLINE
DEGRADED
OFFLINE
RECOVERING

Cached or previously synchronized information may remain available subject to freshness policy.

46. ARIA INTEGRATION

Aria shall access ArcGIS through structured OneDraft geospatial services.

```
USER
↓
ARIA
↓
GEOSPATIAL INTENT
↓
STRUCTURED COMMAND
↓
AUTHORIZATION
↓
SPATIAL SERVICE
↓
ARCGIS ADAPTER
↓
ARCGIS
```

Aria shall not receive unrestricted direct credentials to external GIS systems.

47. ARIA SPATIAL COMMAND MODEL

Spatial commands shall use structured operations.

Conceptually:

```
SpatialCommand
├── action
├── target
├── location
├── dataset
├── filters
├── spatial_operation
├── temporal_state
└── authorization_context
```

This permits deterministic execution and validation.

48. DETERMINISTIC GIS EXECUTION

AI interpretation shall terminate before deterministic GIS execution.

```
NATURAL LANGUAGE
  ↓
ARIA
  ↓
STRUCTURED COMMAND
  ↓
VALIDATION
  ↓
GIS OPERATION
  ↓
DETERMINISTIC RESULT
```

The result shall then be returned to Aria for interpretation or presentation.

49. API ARCHITECTURE

Stack 53.0 shall expose application APIs through the existing OneDraft API Gateway.

Example resources include:

```
GET    /spatial/datasets
GET    /spatial/datasets/{id}
GET    /spatial/layers
GET    /spatial/features
GET    /spatial/entities/{id}
GET    /spatial/search
POST   /spatial/imports
POST   /spatial/sync
POST   /projects/{id}/georeference
POST   /projects/{id}/spatial-context
```

Exact routes shall be formalized through the OneDraft OpenAPI contract.

50. EVENT ARCHITECTURE

Spatial integration shall publish events through the OneDraft Event Bus.

Events shall include:

```
ArcGISServiceRegistered
SpatialDatasetRegistered
```

SpatialLayerRegistered
SpatialImportStarted
SpatialImportCompleted
SpatialImportFailed
SpatialFeatureUpdated
SpatialSynchronizationStarted
SpatialSynchronizationCompleted
SpatialConflictDetected
SpatialRepositoryUpdated
ProjectGeoreferenced

Events shall be versioned.

51. JOB ARCHITECTURE

Long-running ArcGIS operations shall execute through the OneDraft Job & Worker system.

Jobs shall include:

Dataset imports

Bulk feature retrieval

Spatial synchronization

Large-area queries

Spatial indexing

Repository reconciliation

Map generation

Bulk write-back

Each job shall provide:

Job ID

Status

Progress

Retry state

Error state

Result

52. QUEUE ARCHITECTURE

Spatial jobs shall enter appropriate queues.

API
↓
JOB
↓
QUEUE
↓
WORKER
↓
ARCGIS
↓
RESULT
↓
EVENT

Queue isolation shall prevent large GIS operations from blocking ordinary OneDraft application requests.

53. RETRY POLICY

Transient external failures shall be retried according to controlled policy.

Retryable conditions may include:

Temporary network failure

Service timeout

Rate limiting

Temporary provider unavailability

Non-retryable conditions shall terminate immediately with explicit failure state.

54. RATE LIMITING

The ArcGIS integration shall respect provider-specific request limits.

The adapter shall implement:

Request throttling

Backoff

Concurrency limits

Batching

Retry-after handling where provided

Provider limits shall not be circumvented.

55. AUTHENTICATION

External GIS authentication shall be isolated within the integration layer.

Credentials or tokens shall be stored using the OneDraft secrets architecture.

The application shall support the authentication mechanisms permitted by the selected ArcGIS deployment and subscription configuration.

Credentials shall never be persisted in ordinary project records.

56. AUTHORIZATION

ArcGIS operations shall be authorized at multiple levels:

USER
↓
ORGANIZATION
↓
PROJECT
↓
DATASET
↓
OPERATION
↓
ARCGIS

Authorization shall be evaluated before external modification.

57. TENANT ISOLATION

External GIS information associated with one organization shall not automatically become available to another organization.

The spatial repository shall retain tenant context where applicable.

Global geographic indexing shall not override project or organization access boundaries.

58. DATA CLASSIFICATION

Spatial information shall be classified before repository contribution.

Possible classifications include:

PUBLIC
LICENSED
ORGANIZATION
PROJECT
RESTRICTED
PRIVATE

Classification shall govern storage, display, synchronization and write-back.

59. REPOSITORY WRITE AUTHORIZATION

The global repository shall require an explicit authorization pathway.

PROJECT
↓
CLASSIFICATION
↓
OWNER / AUTHORITY
↓
WRITE AUTHORIZATION
↓
VALIDATION
↓
REPOSITORY

This ensures that project information does not become globally visible merely because it exists inside OneDraft.

60. DATA RETENTION

External spatial data shall be retained according to:

Source requirements

License

Project requirements

OneDraft retention policy

Regulatory requirements

Operational requirements

Retention shall be recorded in metadata.

61. LICENSE MANAGEMENT

The Dataset Registry shall retain applicable licensing information.

The system shall distinguish:

Source access

Internal use

Derived data

Redistribution rights

Repository publication

No dataset shall be redistributed beyond its permitted rights.

62. EXTERNAL DATA AUTHORITY

The system shall distinguish:

SOURCE AUTHORITY

from:

ONEDRAFT DERIVED DATA

and:

ONEDRAFT ENGINEERING DATA

A source may be authoritative for its own dataset while OneDraft remains authoritative for its own project model.

63. DATA LINEAGE

The complete lineage shall be:

ARCGIS SOURCE

↓

DATASET

↓

LAYER

↓
FEATURE
↓
INGESTION
↓
TRANSFORMATION
↓
ONEDRAFT ENTITY
↓
PROJECT
↓
ENGINEERING
↓
DOCUMENTATION

This lineage shall be persisted where required.

64. REPOSITORY RECONCILIATION

The reconciliation service shall periodically compare:

EXTERNAL STATE
+
ONEDRAFT STATE
↓
DIFFERENCE MODEL
↓
RESOLUTION
↓
SYNCHRONIZED STATE

Reconciliation shall be observable and auditable.

65. GLOBAL REPOSITORY GROWTH

The implementation shall permit incremental expansion.

PROJECT 1
↓
PROJECT 2
↓
PROJECT 3
↓
CITY
↓
REGION
↓
COUNTRY
↓
CONTINENT
↓

GLOBAL

The infrastructure shall not require a complete planetary dataset before operation begins.

66. EXISTING-WORLD RECONSTRUCTION

Where authorized source information exists, OneDraft may progressively reconstruct representations of existing built environments.

The process shall be:

```
SOURCE DATA
↓
GEOGRAPHIC RECORD
↓
STRUCTURE PLACEHOLDER
↓
ADDITIONAL DATA
↓
GEOMETRIC MODEL
↓
ENGINEERING RECORD
```

The resulting information shall preserve the distinction between observed source information and inferred or newly generated information.

67. NEW-WORLD CONTRIBUTION

New OneDraft projects may contribute appropriately authorized information.

```
NEW PROJECT
↓
ENGINEERING MODEL
↓
VALIDATION
↓
ACCEPTANCE
↓
GEOGRAPHIC EXTRACTION
↓
GLOBAL REPOSITORY
```

This creates a mechanism through which new engineering information can progressively enrich the geographic repository.

68. OLD + NEW CONVERGENCE

The implementation establishes the operational mechanism by which:

PRE-EXISTING WORLD
+
NEW ONEDRAFT ENGINEERING
↓
COMMON SPATIAL FRAMEWORK
↓
CONTINUOUS WORLD MODEL

Existing information and newly generated information remain distinguishable while becoming spatially related.

69. CITY IMPLEMENTATION

The first practical expansion target may operate at city scale.

CITY
├── EXISTING STRUCTURES
├── EXISTING INFRASTRUCTURE
├── REGISTERED PROJECTS
├── ONEDRAFT PROJECTS
└── EXTERNAL GIS DATA

The same implementation can subsequently scale outward.

70. GLOBAL IMPLEMENTATION

At scale:

CITY
↓
COUNTRY
↓
CONTINENT
↓
PLANET

The repository becomes a distributed geographic information environment rather than a collection of isolated project databases.

71. EARTH-DATA EXTENSIBILITY

The ArcGIS implementation shall not be restricted to architectural structures.

The spatial provider and canonical model shall remain extensible to:

Land

Water

Infrastructure

Transportation

Ecology

Environmental datasets

Humanitarian geographic datasets

Future Earth-data systems

Architectural information shall remain the initial application domain.

72. FUTURE DISPLACEMENT DATA INTERFACE

The repository architecture shall be capable of accepting future aggregated geographic movement information.

LOCATION A
↓
MOVEMENT EVENT
↓
LOCATION B
↓
TIME

Future implementations shall apply appropriate privacy, aggregation and authorization controls.

73. FUTURE ECOLOGICAL DATA INTERFACE

Future ecological information may be spatially associated with built-environment data.

STRUCTURES
+

LAND
+
WATER
+
ECOLOGY
+
TIME

The ArcGIS integration shall therefore become an extensible spatial infrastructure layer rather than a building-only service.

74. MAP APPLICATION IMPLEMENTATION

The frontend shall consume OneDraft spatial APIs.

FRONTEND
↓
MAP CONTROLLER
↓
SPATIAL API
↓
SPATIAL INDEX
↓
CANONICAL DATA

The frontend shall not directly expose unrestricted external GIS credentials.

75. MAP LAYER MODEL

Map layers shall be composable.

BASE MAP
+
BOUNDARIES
+
INFRASTRUCTURE
+
STRUCTURES
+
PROJECTS
+
SITE
+
ENGINEERING

Layer visibility shall be governed by authorization and project context.

76. PROJECT GIS WORKSPACE

OneDraft shall provide a GIS-aware project workspace capable of displaying:

Project location

Site boundary

Existing structures

External GIS layers

Project geometry

Engineering context

Spatial relationships

The workspace shall connect geographic context with architectural design.

77. CAD / GIS COORDINATION

The implementation shall provide a controlled bridge:

GIS
↓
SITE CONTEXT
↓
PROJECT COORDINATES
↓
CAD / BIM

and:

CAD / BIM
↓
PROJECT GEOMETRY
↓
GEOREFERENCE
↓
GIS

The transformation chain shall remain traceable.

78. SURVEY INTEGRATION

Survey data may be incorporated through the same spatial pipeline.

SURVEY
↓

IMPORT
↓
COORDINATE VALIDATION
↓
NORMALIZATION
↓
PROJECT SITE
↓
GLOBAL SPATIAL MODEL

Survey information shall retain source metadata.

79. EXISTING CAD / BIM INTEGRATION

Existing CAD and BIM data may be geographically registered.

EXISTING CAD/BIM
↓
IMPORT
↓
GEOREFERENCE
↓
VALIDATION
↓
SPATIAL MODEL
↓
PROJECT CONTEXT

The implementation shall preserve original source identity.

80. PDF / IMAGE GEOSPATIAL CONTEXT

Where source documents contain geographic information, the existing OneDraft import architecture may provide spatial metadata to the geospatial subsystem.

The system shall distinguish:

Source document

Extracted information

Geospatial interpretation

Validated spatial entity

No unverified extraction shall automatically become authoritative spatial state.

81. OBSERVABILITY

The ArcGIS implementation shall emit telemetry for:

Request count

Request latency

Provider errors

Authentication failures

Dataset imports

Feature imports

Synchronization

Write-back

Conflicts

Retries

Queue depth

Worker utilization

Spatial query performance

82. AUDIT LOGGING

Significant operations shall be auditable.

Audit events shall include:

Who

What

When

Dataset

Project

Operation

Result

Source

Destination

Authorization context

Audit records shall be protected from unauthorized modification.

83. TESTING ARCHITECTURE

Stack 53.0 shall require:

Unit tests

Adapter tests

GIS contract tests

Integration tests

Spatial geometry tests

CRS transformation tests

Schema mapping tests

Synchronization tests

Conflict tests

Security tests

Tenant-isolation tests

Performance tests

End-to-end tests

84. MOCK ARCGIS PROVIDER

Development and automated testing shall support a mock GIS provider.

OneDraft

↓

Provider Interface

↓

Mock ArcGIS Provider

This permits testing without requiring live external GIS infrastructure for every development operation.

85. INTEGRATION ENVIRONMENT

A controlled integration environment shall support live ArcGIS testing.

The environment shall separate:

DEVELOPMENT
STAGING
PRODUCTION

External credentials and datasets shall remain environment-specific.

86. DEPLOYMENT ARCHITECTURE

The ArcGIS integration shall deploy as part of the OneDraft application infrastructure.

Conceptually:

```
WEB
|
API
|
SPATIAL SERVICE
|
QUEUE
|
WORKER
|
ARCGIS ADAPTER
|
ARCGIS
```

Supporting services shall remain integrated with PostgreSQL, object storage, observability and secrets management.

87. CONFIGURATION

ArcGIS configuration shall be environment-controlled.

Configuration shall include:

Service endpoints

Provider settings

Timeouts

Retry policies

Concurrency

Dataset policies

Synchronization schedules

Feature flags

Secrets shall remain outside source code.

88. SECURITY MODEL

The implementation shall preserve:

Authentication

Authorization

Tenant isolation

Secret management

Auditability

Data classification

Provider access boundaries

Project access boundaries

The GIS connector shall be treated as an external integration boundary.

89. FAILURE MODES

The system shall explicitly handle:

SERVICE UNAVAILABLE
AUTHENTICATION FAILURE
AUTHORIZATION FAILURE
TIMEOUT
RATE LIMIT
INVALID RESPONSE
INVALID GEOMETRY
SCHEMA CONFLICT
CRS FAILURE
DATA CONFLICT
DATABASE FAILURE
QUEUE FAILURE

Each condition shall produce a defined operational response.

90. RECOVERY

Recoverable operations shall support:

Retry

Resume

Reconciliation

Rollback where applicable

Dead-letter processing

Operator review

The system shall preserve enough state to determine what occurred during an interrupted operation.

91. DATA INTEGRITY

The implementation shall preserve transactional integrity between:

Spatial entity

Dataset

Layer

Project

Provenance

Version

Relationship

Partial imports shall not create falsely complete canonical records.

92. TRANSACTION BOUNDARIES

Database transactions shall be defined around logically atomic operations.

External ArcGIS calls shall not be assumed to participate in OneDraft database transactions.

The implementation shall therefore use explicit synchronization state.

93. IDEMPOTENCY

Import and synchronization operations shall be idempotent where practical.

Repeated execution of the same operation shall not create uncontrolled duplicate spatial entities.

External identifiers and synchronization state shall support idempotent processing.

94. REPOSITORY CONSISTENCY

The system shall distinguish:

SOURCE CONSISTENCY

from:

ONEDRAFT CONSISTENCY

from:

SYNCHRONIZATION CONSISTENCY

The system shall expose synchronization state rather than implying that all systems are simultaneously identical.

95. VERSION CONTROL

Implementation artifacts shall be version controlled.

This includes:

Adapter code

Schemas

API contracts

Dataset mappings

Transformation rules

Configuration

Database migrations

Synchronization logic

Tests

96. MIGRATION ARCHITECTURE

Spatial database changes shall be introduced through controlled migrations.

Migrations shall support:

Forward migration

Validation

Rollback strategy where feasible

Version tracking

No manual production database modification shall be treated as the standard deployment mechanism.

97. API CONTRACT

The OneDraft OpenAPI architecture shall formally define all public and internal spatial endpoints.

Contracts shall include:

Request schemas

Response schemas

Authentication

Authorization

Errors

Pagination

Filtering

Spatial parameters

Versioning

98. PAGINATION

Large ArcGIS result sets shall use controlled pagination or batching.

The implementation shall prevent unrestricted feature retrieval from exhausting application resources.

Pagination state shall remain deterministic where the external provider supports appropriate ordering.

99. BULK PROCESSING

Large geographic datasets shall be processed through asynchronous bulk jobs.

DATASET
↓
PARTITION
↓
QUEUE
↓
WORKERS
↓
VALIDATE
↓
PERSIST
↓
INDEX

Workers shall be horizontally scalable.

100. PERFORMANCE MODEL

Performance targets shall be established for:

Spatial query latency

Dataset ingestion

Feature processing

Synchronization

Map rendering

Project context retrieval

Global search

Performance shall be measured rather than assumed.

101. SCALABILITY MODEL

The architecture shall scale independently across:

API
WORKERS
QUEUES
DATABASE
SPATIAL INDEX
OBJECT STORAGE

The ArcGIS adapter shall not become a single system-wide bottleneck.

102. GLOBAL REPOSITORY PARTITIONING

As repository volume increases, spatial data may be partitioned by:

Geography

Tenant

Dataset

Temporal range

Entity class

The partitioning strategy shall remain transparent to the canonical API.

103. GLOBAL INDEXING

Global spatial indexing shall support geographic discovery without requiring full-table scans of the entire repository.

Index design shall account for:

Geometry

Location

Administrative hierarchy

Entity type

Temporal state

Project

Dataset

104. REPOSITORY ARCHIVE

Superseded spatial data may be moved into archival storage while preserving:

Identity

Provenance

Version

Temporal state

Retrieval path

Archiving shall not destroy required lineage.

105. REPOSITORY BACKUP

Spatial information shall participate in the existing OneDraft backup architecture.

Backups shall include appropriate:

Database records

Spatial metadata

Repository configuration

Object artifacts

Provenance

Recovery shall be tested according to the existing disaster-recovery architecture.

106. ARCGIS SERVICE HEALTH

The integration shall periodically assess configured service health.

Health state may be:

AVAILABLE

DEGRADED

UNAVAILABLE

AUTHENTICATION_ERROR

CONFIGURATION_ERROR

Service health shall be exposed through OneDraft observability.

107. DATASET HEALTH

Dataset health shall include:

Last synchronization

Record count

Validation failures

Schema changes

Source availability

Version

Conflict count

This permits administrators to identify deteriorating external integrations.

108. SCHEMA DRIFT

The system shall detect external schema changes.

```
EXPECTED SCHEMA
+
CURRENT ARCGIS SCHEMA
↓
SCHEMA COMPARISON
↓
COMPATIBLE
OR
DRIFT DETECTED
```

Schema drift shall not silently break canonical mappings.

109. VERSION DRIFT

The synchronization system shall detect external version changes where provider information permits.

Version differences shall become synchronization events.

110. FEATURE DELETION

External feature deletion shall be handled as a temporal state transition rather than automatic destruction of the OneDraft record.

```
EXTERNAL FEATURE
↓
DELETED EXTERNALLY
↓
SYNC EVENT
↓
```

SUPERSEDED / REMOVED STATE

Retention policy shall determine whether historical information remains available.

111. SOURCE REPLACEMENT

If an external dataset is replaced, OneDraft shall preserve the lineage between the old and new dataset where appropriate.

```

DATASET A
  ↓
SUPERSEDED BY
  ↓
DATASET B

```

112. MULTI-SOURCE RECONCILIATION

Where multiple GIS sources represent the same geographic feature, the system shall preserve source distinctions.

```

SOURCE A
  \
   → CANONICAL CONTEXT
  /
SOURCE B

```

The system shall not automatically declare one source authoritative unless policy establishes that relationship.

113. SPATIAL DATA CONFIDENCE

The implementation shall maintain confidence metadata.

Possible states include:

```

VERIFIED
SOURCE-CONFIRMED
IMPORTED
DERIVED
ESTIMATED
UNRESOLVED

```

Confidence shall not be confused with legal or regulatory authority.

114. HUMAN REVIEW

Where automated reconciliation cannot safely resolve a conflict, the system shall create a review task.

CONFLICT
↓
REVIEW QUEUE
↓
AUTHORIZED REVIEW
↓
DECISION
↓
RESOLUTION

Human decisions shall be auditable.

115. PROJECT ACCEPTANCE INTEGRATION

Global repository contribution shall occur only according to the project's acceptance state and applicable authorization policy.

DESIGN
↓
VALIDATION
↓
REVIEW
↓
ACCEPTANCE
↓
ELIGIBLE DATA
↓
GLOBAL CONTRIBUTION

This protects the global repository from incomplete project states.

116. ENGINEERING RECORD CONTRIBUTION

Validated engineering information may be extracted into geographic context.

Examples include:

Building footprint

Building height

Levels

Site boundary

Structure classification

Construction state

Project date

Spatial relationships

The system shall preserve source provenance for each contribution.

117. WORLD-MODEL UPDATE

A repository contribution shall update the world model only through controlled state transition.

```
PROJECT DATA
↓
VALIDATION
↓
CLASSIFICATION
↓
AUTHORIZED CONTRIBUTION
↓
WORLD MODEL
```

The world model shall therefore evolve through explicit information events.

118. FUTURE PROJECT CONTEXT

Future OneDraft projects may query the repository to retrieve relevant existing information.

```
NEW PROJECT
↓
SITE
↓
GLOBAL SEARCH
↓
EXISTING CONTEXT
↓
PROJECT MODEL
```

This creates cumulative reuse of geographic engineering information.

119. ENGINEERING KNOWLEDGE CONTINUITY

The implementation establishes:

PAST ENGINEERING
↓
GLOBAL REPOSITORY
↓
CURRENT ENGINEERING
↓
FUTURE ENGINEERING

Information does not need to be recreated independently for every future project when authorized reusable information already exists.

120. APPLICATION IMPLEMENTATION PHASES

The implementation shall proceed through controlled phases.

PHASE 1
ARCGIS CONNECTIVITY
↓
PHASE 2
DATASET / LAYER REGISTRY
↓
PHASE 3
READ / QUERY SERVICES
↓
PHASE 4
INGESTION / NORMALIZATION
↓
PHASE 5
SPATIAL PERSISTENCE
↓
PHASE 6
PROJECT GEOREFERENCING
↓
PHASE 7
SYNCHRONIZATION
↓
PHASE 8
AUTHORIZED WRITE-BACK
↓
PHASE 9
GLOBAL REPOSITORY
↓
PHASE 10
WORLD-MODEL APPLICATION

121. MINIMUM VIABLE IMPLEMENTATION

The first executable implementation shall establish:

ArcGIS provider adapter

Service registration

Dataset registry

Layer registry

Feature retrieval

Spatial normalization

PostgreSQL spatial persistence

Project georeferencing

Spatial search

Provenance

Authentication

Authorization

Observability

Testing

This establishes a functioning OneDraft–ArcGIS integration without requiring the complete planetary repository on day one.

122. EXPANDED IMPLEMENTATION

Following the MVP integration, the system shall expand to:

Bulk ingestion

Synchronization

Versioning

Conflict resolution

Global project index

Structure registry

Write-back

Temporal world model

Global mapping

Large-scale spatial analysis

123. IMPLEMENTATION READINESS

Stack 53.0 shall be considered implementation-ready when the following are defined:

DOMAIN MODEL
+
DATABASE MODEL
+
API CONTRACT
+
ARCGIS ADAPTER
+
AUTHENTICATION
+
AUTHORIZATION
+
DATA MAPPING
+
INGESTION
+
SYNCHRONIZATION
+
WRITE-BACK
+
PROVENANCE
+
TESTING
+
DEPLOYMENT

The resulting specification shall be sufficient to begin implementation without requiring a new conceptual architecture.

124. END-TO-END IMPLEMENTATION WORKFLOW

The complete implementation workflow shall be:

ARCGIS SERVICE

↓
DISCOVERY
↓
DATASET REGISTRATION
↓
LAYER REGISTRATION
↓
FEATURE QUERY
↓
INGESTION
↓
NORMALIZATION
↓
CRS TRANSFORMATION
↓
VALIDATION
↓
POSTGRESQL
↓
SPATIAL INDEX
↓
PROJECT CONTEXT
↓
ARIA
↓
ENGINEERING
↓
VALIDATION
↓
ACCEPTANCE
↓
AUTHORIZED WRITE-BACK
↓
GLOBAL REPOSITORY
↓
WORLD MODEL

125. INTEGRATION WITH 1.0–52.0

The OneDraft architecture now progresses:

1.0–5.0
FOUNDATION
↓
6.0–16.0
EXECUTABLE APPLICATION FOUNDATION
↓
17.0–22.0
CORE COMPUTATIONAL & DOCUMENTATION ENGINES
↓
23.0–29.0
RUNTIME, JOBS, ARTIFACTS, EVENTS & SECURITY
↓
30.0–36.0
OPERATIONS, QUALITY, INFRASTRUCTURE & DATA GOVERNANCE

↓
37.0–40.0
AI, CAD STANDARDS & DATA EXCHANGE
↓
41.0–49.0
USER PLATFORM, LIFECYCLE, COMMERCIAL & SCALE SYSTEMS
↓
50.0
END-TO-END SYSTEM INTEGRATION
↓
51.0
GLOBAL GEOSPATIAL INTELLIGENCE, MAPPING &
EARTH DATA INTEGRATION
↓
52.0
GLOBAL GEOSPATIAL APPLICATION, REPOSITORY &
WORLD-MODEL IMPLEMENTATION
↓
53.0
ONEDRAFT-ARCGIS REPOSITORY APPLICATION
IMPLEMENTATION

126. ARCHITECTURAL COMPLETION STATE

Stack 53.0 establishes the executable bridge between OneDraft and the external ArcGIS ecosystem.

The architecture now provides:

ONEDRAFT
+
ARIA
+
CAD / BIM
+
GEOMETRY
+
PROJECT MODEL
+
GLOBAL SPATIAL MODEL
+
ARCGIS CONNECTIVITY
+
GIS DATA INGESTION
+
SPATIAL DATABASE
+
SYNCHRONIZATION
+
PROVENANCE
+
VERSIONING
+
AUTHORIZED WRITE-BACK

+
GLOBAL REPOSITORY

The OneDraft project can therefore operate as both:

AN ENGINEERING PROJECT

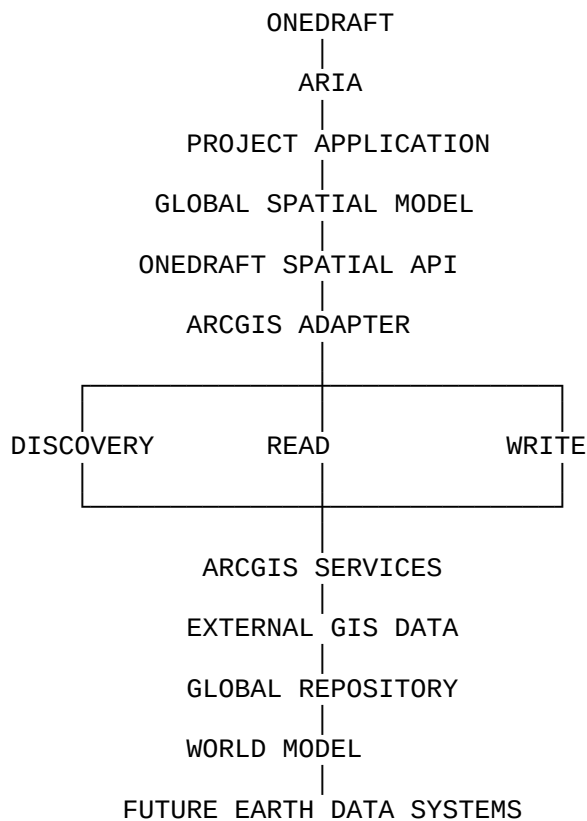
and:

A GEOGRAPHICALLY REGISTERED NODE

within the larger OneDraft spatial repository.

127. MASTER IMPLEMENTATION MODEL

The complete Stack 53.0 architecture is:



128. FINAL IMPLEMENTATION PRINCIPLE

Stack 53.0 establishes the principle:

ARCGIS IS THE EXTERNAL GEOSPATIAL INTERFACE.

ONEDRAFT IS THE ENGINEERING PLATFORM.

THE ADAPTER IS THE CONTROLLED BRIDGE.

THE SPATIAL MODEL IS THE CANONICAL ONEDRAFT REPRESENTATION.

ARIA IS THE INTELLIGENT ORCHESTRATOR.

VALIDATION IS THE DETERMINISTIC CONTROL.

PROVENANCE PRESERVES SOURCE HISTORY.

AUTHORIZATION CONTROLS REPOSITORY CONTRIBUTION.

THE GLOBAL REPOSITORY PROVIDES CONTINUITY.

THE WORLD MODEL PROVIDES TEMPORAL AND SPATIAL CONTEXT.

The implementation therefore establishes the foundation through which OneDraft may progressively connect architectural projects, existing structures, GIS datasets and future geographic information into a persistent global engineering information environment.

129. COMPLETION STATUS

Stack: 53.0

System: OneDraft CAD-AI

Intelligence Layer: Aria

Primary Function: OneDraft–ArcGIS Repository Application Implementation

Primary External Integration: ArcGIS

Primary Internal Integration: OneDraft Global Spatial Model

Primary Persistence: OneDraft PostgreSQL Spatial Repository

Primary Application Pattern: Provider Adapter + Canonical Spatial Model

Primary Data Flow: ArcGIS → OneDraft

Authorized Reverse Flow: OneDraft → ArcGIS

Spatial Scope: Project → Site → Structure → City → Region → Country → Continent → Global

Temporal Scope: Historical → Existing → Current → Proposed → Future

Implementation Scope: Connectivity → Ingestion → Normalization → Persistence → Synchronization → Write-Back → Global Repository

Architectural Status: IMPLEMENTATION SPECIFICATION